

JaiSheng Ong

(513) 918-9488 | ongjs@mail.uc.edu | [linkedin.com/in/jaisheng-ong](https://www.linkedin.com/in/jaisheng-ong) | github.com/jaisheng-ong

EDUCATION

University of Cincinnati

Bachelor of Science, Computer Science — GPA: 3.4

Cincinnati, OH

Expected May 2028

- Relevant Coursework: Data Structures, Discrete Computational Structures

Cincinnati State Technical and Community College

Associate of Applied Science, Software Engineering Technology — GPA: 4.0

Cincinnati, OH

Graduated: Spring 2024

- Relevant Coursework: System Analysis & Design, Python Object-Oriented Programming

EXPERIENCE

Machine Learning Researcher

Jan 2026 – May 2026

Ming Chi University of Technology | Internship

Taiwan

- **Architected an end-to-end machine learning pipeline** in Python to predict highly volatile industrial wastewater metrics.
- **Engineered advanced time-series features** by implementing a **2D-state Kalman filter** and **ARIMA** one-step-ahead forecasting, effectively de-noising sensor volatility and completely eliminating temporal data leakage.
- **Conducted Bayesian hyperparameter optimization** via **Optuna** to tune a deep learning **LSTM architecture**, actively comparing sequence-learning capacity against traditional tree-based models.
- **Optimized for production viability**, demonstrating that human-guided mathematical forecasting paired with Extra Trees was 3x more computationally efficient and more stochastically stable than deep learning alternatives for small-scale edge deployment.
- **Wrote an IEEE-formatted scientific paper** detailing the ablation study, feature engineering progression, and computational cost analysis.

Software Engineer

May 2025 – Aug 2025

Aubot | Internship

Remote

- **Reduced lesson publishing time by 30%** by auditing and correcting 120+ technical, grammar, and clarity errors across 10+ advanced **Python** course documents.
- **Strengthened content reliability** by authoring automated test scripts to catch incorrect solution methods and contributing targeted fixes via **Git** pull requests across a collaborative codebase.

PROJECTS

Ski and Snowboard Rental System | Python, OOP

- Architected a full-featured rental management system in **Python** with OOP design to support multi-rate bookings (hourly, daily, weekly) and real-time inventory browsing.
- Reduced manual pricing errors by implementing automated discount logic for family rentals and coupon code validation at point of rental.

Passenger & Employee Management System | Visual Basic, MySQL

- Built a dual-role authentication system in **Visual Basic** backed by **MySQL**, enabling secure, validated login for both passenger and employee account types.
- Hardened application security by enforcing role-based access control and input validation, eliminating unauthorized cross-role access between user tiers.

TECHNICAL SKILLS

Languages: Python, C++, VB, VBA, Assembly, MatLab, NI LabVIEW, MySQL, Bash

Technologies: Git, GitHub, Linux, LaTeX, Jupyter

Miscellaneous: Mandarin Chinese fluency

Available for Co-op/Internship: **Fall 2026**